

$$= \frac{2(4)^{\frac{5}{2}}}{5} - \frac{2(4)^{\frac{3}{2}}}{3} + 2(4)^{\frac{1}{2}} - \underbrace{\left(\frac{2(0)^{\frac{5}{2}}}{5} - \frac{2(0)^{\frac{3}{2}}}{3} + 2(0)^{\frac{1}{2}} \right)}_0$$

$$= \frac{2(\sqrt{4})^5}{5} - \frac{2(\sqrt{4})^3}{3} + 2\sqrt{4}$$

$$= \frac{2(2)^5}{5} - \frac{2(2)^3}{3} + 2(2)$$

$$= \frac{64 \times 3}{5 \times 3} - \frac{16 \times 5}{3 \times 5} + \frac{4 \times 15}{1 \times 15}$$

$$= \frac{192 - 80 + 60}{15} = \frac{172}{15} \quad \#$$

c) $\int \sin^4 x \cos x \, dx$

гн $u = \sin x$

$du = \cos x \, dx$

$dx = \frac{du}{\cos x}$

$\int \sin^4 x \cos x \, dx$

$= \int u^4 \cancel{\cos x} \frac{du}{\cancel{\cos x}}$

$= \int u^4 \, du = \frac{u^5}{5} + C = \frac{\sin^5 x}{5} + C$

d) $\int \frac{x+3}{(x^2+6x)^2} \, dx$

гн $u = x^2 + 6x$

$du = (2x+6) \, dx$

$dx = \frac{du}{2x+6}$

$\int \frac{x+3}{u^2} \frac{du}{2x+6}$

$= \int \frac{\cancel{x+3}}{u^2} \frac{du}{2\cancel{(x+3)}}$

$= \frac{1}{2} \int \frac{1}{u^2} \, du$

$= \frac{1}{2} \int u^{-2} \, du$

$= \frac{1}{2} \left(\frac{u^{-1}}{-1} \right) + C$

$= -\frac{1}{2} (x^2+6x)^{-1} + C$

4) เนื่องจาก $\int \sec x \tan x \, dx = \sec x + C$

ดังนั้น $\int \sec(\pi x) \tan(\pi x) \, dx = \frac{1}{\pi} \sec(\pi x) + C$
 $a = \pi \quad b = 0$

5) เนื่องจาก $\int \frac{1}{2\sqrt{x}} \, dx = \int \frac{1}{2x^{\frac{1}{2}}} \, dx = \frac{1}{2} \int x^{-\frac{1}{2}} \, dx = \frac{1}{2} \left(\frac{x^{\frac{1}{2}}}{\frac{1}{2}} \right) + C = \sqrt{x} + C$

ดังนั้น $\int \frac{1}{2\sqrt{3x+4}} \, dx = \frac{1}{3} \left(\sqrt{3x+4} \right) + C$
 $a = 3 \quad b = 4$

6) เนื่องจาก $\int \frac{1}{x^2 + 4} \, dx = \int \frac{1}{x^2 + 2^2} \, dx = \frac{1}{2} \arctan\left(\frac{x}{2}\right) + C$

ดังนั้น $\int \frac{1}{(2x+1)^2 + 4} \, dx = \frac{1}{2} \left(\frac{1}{2} \arctan\left(\frac{2x+1}{2}\right) \right) + C$
 $a = 2 \quad b = 1$
 $= \frac{1}{4} \arctan\left(\frac{2x+1}{2}\right) + C$

7) เนื่องจาก $\int \frac{1}{x} \, dx = \ln|x| + C$

ดังนั้น $\int_1^2 \frac{1}{2-3x} \, dx = \left(-\frac{1}{3} \ln|2-3x| \right) \Big|_1^2$
 $= -\frac{1}{3} (\ln|2-3(2)| - \ln|2-3(1)|)$
 $= -\frac{1}{3} (\ln|-4| - \ln|-1|)$
 $= -\frac{1}{3} (\ln 4 - \ln 1)$

$$= \frac{\ln 4}{3}$$

$$= \frac{\ln 2^2}{3}$$

$$= \frac{2 \ln 2}{3}$$

$$\int \frac{f'(x)}{f(x)} dx = \ln|f(x)| + C$$

แบบฝึกหัดที่ 8 หน้า 8

1) $\int \frac{2x}{x^2 + 4} dx = \dots\dots\dots$

2) $\int \frac{x^2 + 1}{x^3 + 3x - 4} dx = \dots\dots\dots$
 $= \dots\dots\dots$

3) $\int \frac{1}{x \ln x} dx = \dots\dots\dots$

4) $\int \frac{e^x}{e^x + 3} dx = \dots\dots\dots$

5) $\int \frac{\sin x}{1 + \cos x} dx = \dots\dots\dots$
 $= \dots\dots\dots$